

PROJECT AREA: MT. GUNSON PROJECT No. 023
 SPUDDED: 21-3-73 COMPLETED: 28-3-73
 LOCATION: CATTLEGRID CO-ORDS: X: 6518799.02m N Y: 704002.57m E
 ELEVATION: 89.22m DIRECTION: INCLINATION: 90° TOTAL DEPTH: 44.21m
 HOLE TYPE: ROTARY & DIAMOND DRILLING CONTRACTOR: AFRAC DRILLER: R. TAYLOR; W. BLANCHE
 LOGGED BY: R. TURNBULL DATE: 29-3-73 BIT SIZE: 178mm TO 6.1m; 121mm TO 33.07; NXYT TO 44.21m
K. MAIDEN

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE			
							Cu			
NON-CORING	0	UNCONSOLIDATED SEDIMENTS 0-9.1m		0						
					76010	3.0	15			
				3.0						
					76011	3.1	10			
	5			6.1						
					76012	3.0	15			
				9.1						
	10	WHYALLA SANDSTONE 9.1-37.57 9.1-21.3: Pink to grey Lss, damp; Trace of weathered silicified Lss at top			76013	3.1	30			
				12.2						
					76014	3.0	380			
	15			15.2						
					76015	3.1	30			
				18.3						
					76016	3.0	20			

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HOLE No. P300-000

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CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE					
							Cu	Ag	Pb	Zn	Co	Bi
NON-CORING	20											
		21.3 - 33.07: Pink-grey clayey LSs; Wet below 21.3 m.			21.3							
					76017	3.1	15					
					24.4							
					76018	3.0	20					
	25				27.4							
					76019	3.1	10					
					30.5							
					76020	2.57	10					
	30				33.07							
	33.07	33.07 - 36.57: Coarse-grained loosely consolidated LSs & little clay. No visible sulphide.			76021	1.93	20	<2	40	15	5	
	64%				35.0							
	34.49				765	1.0	10	<2	20	40	5	
	63%				36.0							
	35.40				766	1.0	5	<2	60	80	10	<20
	100%	36.57 - 37.57: Clayey LSs, slumped bedding, scattered rounded QTE frags up to 10mm, Minor disseminated sulphide.			37.0							
	37.08				767	1.0	15	<2	160	120	5	<20
	94%	TRANSITION ZONE 37.57 - 38.85			38.0							
	38.41	37.57 - 37.91: LSs & large angular QTE frags.	Minor sulphide		768	1.0	.34	<2	200	120	15	<20
	100%	37.91 - 38.85: Brecciated QTE & sandy matrix, grading into unbrecciated QTE			39.0							
	40.19				769	1.0	.73	5	1000	600	5	<20
	100%	PANDURRA FORMATION 38.85 - 44.21			40.0							
	41.44	38.85 - 39.33: Pink QTE & occasional vughs partly filled by sulphide. Minor sulphide			770	1.0	.12	<2	140	150	5	35
		39.33 - 40.20: QTE breccia; sand, clay &			41.0							
					771	1.0	550	<2	40	40	5	<20
					42.0							

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795 6.1 240-30

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HOLE No. P300-100
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PROJECT AREA: MT. GUNSON

PROJECT No. 402

SPUDED: 7-4-73

COMPLETED: 12-4-73

LOCATION: CATTLEGRID

CO-ORDS: X: 6518793.10m N: 704094.80m E

ELEVATION: 85.57m DIRECTION:

INCLINATION: 90° TOTAL DEPTH: 39.44m

HOLE TYPE: ROTARY & DIAMOND

DRILLING CONTRACTOR: AFRAC

DRILLER: R. TAYLOR; W. BLANCHE

LOGGED BY: A. TURNBULL
K. MATDEN

DATE: 12-4-73 BIT SIZE: 178 mm TO 6.1 m; 121 mm TO 30.76 m; NXTT TO 39.44 m

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 100	SAMPLE No.	Assayed length	ASSAY VALUE			
							Cu			
NON - CORING	0	UNCONSOLIDATED SEDIMENTS 0 - 8m		0						
					75185	3.0	30			
				3.0						
					75186	3.1	15			
	5			6.1						
		WHYALLA SANDSTONE 8.0 - 30.5 m			75187	3.0	240			
		8.0 - 10.4 m: Silicified LSs, weathered.		9.1						
	10	10.4 - 18.3: Yellow LSs, damp.			75188	3.1	50			
				12.2						
					75189	3.0	35			
	15	Samples wet below 15 m.		15.2						
					75190	3.1	40			
		18.3 - 30.76: Pink-grey LSs, & clay;		18.3						
					75191	3.0	30			

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HOLE No. P400-100

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PROJECT AREA: MT. GUNSON.

PROJECT No. 402.

SPUDED: 26 / 3 / 73

COMPLETED: 3 / 4 / 73

LOCATION: CATTLE GRID BLOCK P.

CO-ORDS: X: 6518900.50m N: 704099.90m E

ELEVATION: 86.22m DIRECTION: INCLINATION: -90°. TOTAL DEPTH: 40.86.

HOLE TYPE: ROTARY S DIAMOND. DRILLING CONTRACTOR: AFRAC. DRILLER: TAYLOR/BLANKHE.

LOGGED BY: J.M.H. DATE: 5/4/73. BIT SIZE: 17.8 cm TO 3m; 11.8 cm TO 30.92 m; NEXT TO 40.86.

CORE RECOVERY	METRAGE	DESCRIPTION	Core bed'd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE			
							Cu			
	0	SAND (0-10 m.)		0.						
					75077	3.0	15			
				3						
					0783.1	20				
	5			6.1						
					0793.0	230				
		YELLOW - PINK SANDY CLAY.		9.1						
	10	WHYALLA SAND STONE. (10-33.90)			0803.1	250				
		YELLOW - grey.		12.2						
					0813.0	85				
	15	A/A.		15.2						
					0823.1	40				
				18.3						
					0833.0	10				

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100.	SAMPLE No.	Assayed length	ASSAY VALUE					
							Cu	Ag	Pb	Zn	Co	Bi
	20	20 WHYALLA SANDSTONE (CONT.) grey.		20								
				21.3								
		A/A			5084	3.1	15					
				24.4								
		A/A			085	3.0	10					
				27.4								
		A/A.			086	3.52	10					
	30.92	30.92 - 33.00 FLAT BEDDED, GREY, FRIABLE. 30.92 TO 31.90 INT. COMPLETE CORE LOSS. 31.90 TO 32.65 SOME CORE LOSS.		30.92	73898	2.08	40					
58.31	32.65	33.00 TO 33.30 SLUMPED BEDS.		33.0	899	1.0	10					
95.1	33.45	33.30 TO 33.90 SILICIFIED SANDSTONE FRAGMENTS IN SANDSTONE MATRIX. FEW FRAGMENTS QUARTZITE		34.0	900	1.0	25	<2	60	60	5	
106.1	35.40	PANDURRA FM. (33.90 - 40.86)		35.0	901	1.0	30	<2	50	60	5	
160.1	36.55	33.90 TO 34.85 FRAGMENTED ANGULAR QTZITE WITHIN A GRITTY MATRIX.		36.0	902	1.0	500	<2	260	45	5	
		34.85 TO 35.30 EVEN GRAINED QTZITE 35.30 TO 36.60 FRAGMENTS OF QTZITE IN RED CLAY MATRIX.		37.0	903	0.5	980	<2	20	25	5	
99.1	39.34	36.60 TO 39.10 EVEN-GRAINED QTZITE w/ OCCASSIONAL DISSEMINATED SULPHIDES. RARELY AS BLEBS. SHALLOW DIPPING JTS.		38.0	904	0.5	1400	<2	<2	5	5	
		39.10 TO 39.34 ORANGE CLAY ZONE.		39.0	905	0.5	1400	<2	<2	5	5	
91.1	40.86	39.34 TO 40.86 QTZITE w/ SUB VERTICAL JOINTING		40.0	906	0.5	180	<2	200	10	5	
		40.86 END OF HOLE.		40.86	907	1.0	70	<2	100	60	5	
					908	0.86	35	<2	20	20	5	

CORE RECOVERY	METRAGE	DESCRIPTION	Core bed'd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE			
							Cu			
NON-CORING.	20	WHYALLA SANDSTONE (CONT.) GREY.		20	132	3.3.				
				21.3.						
					75133	3.1	180			
				24.4						
	25				134	3.0	240			
				27.4						
					135	3.1	150			
	30	PINK-GREY CLAY BANDS.		30.5						
					136	3.0	90			
				33.5						
	35				137	3.1	50			
				36.6						
					138	3.0	30			
		VERY HARD - LITTLE CLAY.		39.6						
	40.				139	3.1.	85			

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872 9.1 80-50

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PROJECT AREA: MT. GUNSON. PROJECT No. 402.
SPUDDED: 6th APRIL 1973 COMPLETED: 7th APRIL 1973.
LOCATION: CATTLEGAID BLOCK T. CO-ORDS: X: 6518303.50m N Y: 703509.51m E
ELEVATION: 93.34m DIRECTION: INCLINATION: -90° TOTAL DEPTH: 49.99m.
HOLE TYPE: ROTARY & DIAMOND DRILLING CONTRACTOR: AFRAC. DRILLER: TAYLOR / BLANCHÉ.
LOGGED BY: AGT/JMH. DATE: 7/4/73. BIT SIZE: 17.8cm TO 1.5m; 11.8cm TO 37.15; NEXT TO 49.99.

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE			
							Cu			
NON-CORING.	0	SAND. 0 - 4 M.		0						
					75152	3.0	10			
				3						
		WHYALLA SANDSTONE. 4 - 41.90m			153	3.1	25			
	5	RESILICIFIED WHITISH.								
				6.1						
		WEATHERED. YELLOW.			154	3.0	50			
				9.1						
	10				155	3.1	45			
		WHITE TO GREY.		12.2						
					156	3.0	80			
	15	GREY MOIST.		15.2						
					157	3.1	40			
				18.3						

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32.8
6.1 90-80

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TRUE B.H. No. T300-200 0546

HOLE No. T300-200

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PROJECT AREA: MT. GUNSON PROJECT No. 402
 SPUDDED: 5-4-73 COMPLETED: 9-4-73
 LOCATION: CATTLEGRID CO-ORDS: X: 6518303.75m/N: 703709.09mE
 ELEVATION: 98.06m DIRECTION: INCLINATION: 90° TOTAL DEPTH: 44.09m
 HOLE TYPE: ROTARY & DIAMOND DRILLING CONTRACTOR: AFRAC DRILLER: R. TAYLOR; W. BLANCHE
 LOGGED BY: A. TURNBULL DATE: 10-4-73 BIT SIZE: 178mm TO 6.1m; 121mm TO 30.63m; NXTT TO 44.09m
 K. MAIDEN

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1:100	SAMPLE No.	Assayed length	ASSAY VALUE			
							Cu			
NON-CORING	0	WHYALLA SANDSTONE 0 - 37.25 0 - 4.6: Silicified LSS, weathered.		0						
					75142	3.0	35			
				3.0						
					75143	3.1	30			
	5	4.6 - 21.3: Yellow to grey LSS		6.1						
					75144	3.0	20			
				9.1						
					75145	3.1	30			
	10			12.2						
					75146	3.0	40			
				15.2						
					75147	3.1	80			
				18.3						
					75148	3.0	90			
	20									

CORE RECOVERY	METRAGE	DESCRIPTION	Core bedd'g angle and joint spacing	LOG SCALE: 1 : 100	SAMPLE No.	Assayed length	ASSAY VALUE			
							Cu			
NON - CORING	20									
		21.3 - 30.63 : LSS + iron oxides ; samples wet		21.3	75149	3.1	65			
100 %										
100 %										
98 %										
99 %										
100 %										
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023

COMPLETED: 23-2-73

COORDS X6518401.23 m N Y 703591.61 m E

INCLINATION: -90° TOTAL DEPTH: 44.69 m.

DRILLING CONTRACTOR: AFRAC DRILLER: W. BLANCHE

LOGGED BY: P. CREEVY 2-3-73

BIT SIZE: 12/ mm. TO 30.88m. NXTT TO 44.69m.

TO

CORE RECOVERY	METRAGE	DESCRIPTION	Core, bedding angle & joint spacing	LOG	SAMPLE NO	ASSAYED LENGTH	ASSAY VALUE			
				SCALE 1: 100			Ca			
NON-CORING	0	DUNE SAND 0-6.1m. 0-6.1m. Dune sand & silicified ss (float) at bottom		0						
					75722	3.0	20			
				3.0						
					23	3.1	35			
	5									
		WHYALLA Ss. 6.1-37.85m. 6.1-9.1m. Hard silicified ss, used Tricone bit for this section. Dune sand in fractures.		6.1						
					24	3.0	50			
		9.1-15.2m. Yellow Ss, moist d with 0x. Cu. below 12.2m. Blade bit used, but very hard drilling, probably partially silicified ss all way down.		9.1						
	10				25	3.1	80			
				12.2						
					26	3.0	6500			
	15	15.2-18.3m. Grey-white moist Ss.		15.2						
					27	3.1	780			
		18.3-30.8m. very hard grey white Ss, water used below 18.3m, but hole made water below 24.4m, no clay seen.		18.3						
				28	3.0	230				

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NEW B.H. No. T400-100

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HOLE No. T350/100

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CORE RECOVERY	METRAGE	DESCRIPTION	Core bedding angle & joint spacing	LOG SCALE 1:100	SAMPLE NO	ASSAYED LENGTH	ASSAY VALUE					
							Cu	Ag	Pb	Zn	Co	Bi
100%	42.5	43.0-44.69m. Slightly less well fractured pink qtz. than section above. cavities largely filled & clayey sand, prob. some sulphide in infill material, esp. @ 44.25m.	Δ Δ Δ Δ	43.0								
44.69				44.0	73405	1.0	2500	42	290	280	30	
	45			44.69	06	0.69	590	42	230	680	10	

GEOCHEMICAL MUD SAMPLING AND DRILLING,

PERNATTY LAGOON

K.J. MAIDEN
24th September, 1972

DISTRIBUTION LIST

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2. REGIONAL GEOCHEMICAL SURVEY OF PERNATTY LAGOON
3. DETAILED SURVEY OF DOLOMITE ISLANDS AREA
4. INTERPRETATION OF ANOMALIES
5. CONCLUSION

MAPS

1. ✓ DWG NO. 399 : LOCATION OF MUD SAMPLES, PERNATTY LAGOON
(1:50,000)
2. DWG NO. 403 : PERNATTY LAGOON BEDROCK LITHOLOGY (1:50,000)
3. ✓ DWG NO. 404 : GEOLOGICAL CROSS SECTION, EAST LAGOON TO
DOLOMITE ISLANDS
4. ✓ DWG NO. 405 : BEDROCK LITHOLOGY, DOLOMITE ISLANDS (WEST)
(1:5,000)
5. ✓ DWG NO. 406 : BEDROCK LITHOLOGY, DOLOMITE ISLANDS (EAST)
(1:5,000)
6. ✓ DWG NO. 407 : COPPER GEOCHEMICAL PLAN, DOLOMITE ISLANDS
(WEST) (1:5,000)
- ✓ 7. DWG NO. 408 : COPPER GEOCHEMICAL PLAN, DOLOMITE ISLANDS
(EAST) (1:5,000)
- ✓ 8. DWG NO. 409 : ZINC GEOCHEMICAL PLAN, DOLOMITE ISLANDS
(WEST) (1:5,000)
- ✓ 9. DWG NO. 410 : ZINC GEOCHEMICAL PLAN, DOLOMITE ISLANDS
(EAST) (1:5,000)
- ✓ 10. TARGET "TOM" : COPPER GEOCHEMICAL PLAN (1:2,500)
- ✓ 11. TARGET "TOM" : ZINC GEOCHEMICAL PLAN (1:2,500)
- ✓ 12. TARGET "DICK" : COPPER GEOCHEMICAL PLAN (1:2,500)
- ✓ 13. TARGET "DICK" : ZINC GEOCHEMICAL PLAN (1:2,500)
- ✓ 14. TARGET "HARRY" : COPPER GEOCHEMICAL PLAN (1:2,500)
- ✓ 15. TARGET "HARRY" : ZINC GEOCHEMICAL PLAN (1:2,500)

1. SUMMARY AND RECOMMENDATIONS

- 1.1 A regional geochemical mud-sampling survey of Pernatty Lagoon revealed only one area of interest, in the vicinity of the dolomite islands in the north-western part of the lagoon.
- 1.2 A detailed sampling programme in this area showed a number of low order copper anomalies, three of which were designated target areas for drilling.
- 1.3 A D9 bulldozer, hired to put trenches through the lagoon mud to the target areas, reached the first group of anomalies (Target "Tom") but failed to reach the other target areas. The other two target areas remain untested. In any future drilling of Pernatty Lagoon, a more satisfactory method of getting the rig to the drill site will have to be found.
- 1.4 Drilling on Target "Tom" intersected no significant copper mineralisation at the dolomite - red quartzite contact. The dolomite has a high background copper concentration, which accounts for the geochemical anomalies.
- 1.5 In its present form, the geochemical survey method has limited use as an exploration tool. It is the only method available for determining the lithology beneath the lagoon mud. However, geochemical anomalies appear to be merely a reflection of high background copper in the